RELATED WORK AND POSITIONING

TEAM A1 — ROLE 1

Abstract. Where this paper sits, and—stated plainly—which of its results are already known. Two of our five findings have close published antecedents; one is crowded; two appear to be new. We say which is which here rather than let a referee discover it.

1. Curtailable load contracts

The instrument we study is a curtailment tariff, and the theory of selling curtailable service is mature. Chao and Wilson [1987] established that a menu of contingent curtailment contracts induces efficient self-selection and dominates random rationing; the market-structure and investment extensions are Chao et al. [1988]. Oren and Smith [1992] is the canonical operations treatment of designing and dispatching a curtailable-service program, and is the closest existing analogue to our dispatch comparison. Strauss and Oren [1993] adds notification lead time—relevant because we assume the load is instantaneously curtailable, an assumption their results show is not innocuous. Kamat and Oren [2002] price interruptible supply as a compound option and Baldick et al. [2006] derive a retailer's optimal interruption policy under stochastic prices; neither considers emissions, which is the opening we take. In our own venue, Fattahi et al. [2023] is the current reference point for direct load-control contract design, and is cost-focused with no carbon analysis.

Our surplus contract is a degenerate one-class priority contract, and our budget family d_τ is a priority-service menu. We claim no novelty in the contract form. What we add is the carbon ledger over that family and the instrument comparison of §5.3.

2. Flexible computing as a grid resource

That computing load responds to electricity signals at all traces to Qureshi et al. [2009]; Liu et al. [2011] showed geographic load balancing can raise renewable utilization, though under a fixed generation mix—the extension to endogenous renewable capacity is where our capacity-substitution channel lives. Wierman et al. [2014] surveys operational feasibility, and Radovanović et al. [2023] documents a deployed carbon-shifting system at Google, which uses marginal-intensity signals rather than a surplus rule. For mining specifically, Menati et al. [2023b] finds the carbon footprint of mining load varies up to 50% from the system average by location, and Menati et al. [2023a] is the reference ERCOT simulation. van der Jagt et al. [2024] models exactly our K_M object as a "demand sink" and finds low-capital-cost sinks cut curtailment substantially; Riepin and Brown [2024] raises the additionality concern that bears on our Pareto framing.

3. What is already known, stated plainly

Our negative result is not new. Knittel et al. [2025] find, empirically, that flexible data centers flatten net demand, cut system costs, and *raise* emissions when the renewable share is low, with the sign turning favourable as penetration grows. That is our "the unconstrained rule raises carbon except in a cheap-renewable corner," in a different currency. The general mechanism is

Date: Supporting section.

older still: Holland and Mansur [2008] showed that reducing demand variance raises emissions in some regions and lowers it in others. We therefore do not present the direction of the effect as a finding. What we contribute is the *exact analytic boundary* in primitives, where they give regional simulations and reduced-form estimates.

This places our result in a recognized class. Carson and Novan [2013] found bulk storage arbitrage adds roughly 0.19 tons CO_2 per MWh stored in ERCOT, and Hittinger and Azevedo [2015] confirmed the sign nationally while showing it flips as renewable capacity rises—the empirical shadow of a classification condition. Ding et al. [2023] shows curtailment caps can cut renewable capacity and raise CO_2, which is our feedback from surplus use to K_R running in reverse.

Our distribution-free statement is crowded, not pre-empted. We have found no proof of the Pareto claim for all F. But van der Jagt et al. [2024] simulates the mechanism, Butters et al. [2025] gives the full structural equilibrium for a flexible asset in *Econometrica*, and Hittinger and Azevedo [2015] documents the conditional flip. The honest defense is not novelty of conclusion but of scope: distribution-freeness buys robustness to exactly the heterogeneity that Novan [2015], Callaway et al. [2018], and Menati et al. [2023b] document empirically— identical resources, wildly different carbon signs by region and season.

Our fossil-minimizing dispatch result is short and, we suspect, folk knowledge. It is a lemma, and we present it as one.

4. Capacity investment

Our three-capacity program is a peak-load-pricing and reliability-constrained capacity model in the tradition of Crew and Kleindorfer [1976], surveyed for operations by Van Mieghem [2003]. The distribution-free cutoffs are newsvendor critical fractiles: $\varphi^\circ = (\beta+p-c-C_F)/(\beta+p-c)$ *is* a critical fractile, and we frame it as one. Hu et al. [2015] warns that data granularity on intermittency drives renewable over-investment—a caution about our own coarse residual-demand primitive. Aflaki and Netessine [2017] is the closest precedent for a perverse-sign result in this literature: carbon pricing can discourage renewable investment under intermittency. That paper is our evidence that "green policy, wrong sign" theorems publish in operations. Kök et al. [2018] is the nearest *Management Science* paper to our object— pricing policy to renewable investment to emissions, with a sign that flips on market characteristics—and Kök et al. [2020] establishes renewable/flexible complementarity. Peng et al. [2024] asks our complementarity question for the (renewable, flexible, storage) triple; since our third asset is a flexible *load* rather than storage, we owe a referee an explicit distinction, and the honest answer is that storage is absent from our model and is the first extension a reader should demand. Sunar and Birge [2019] and Drake et al. [2016] set the modelling conventions we follow.

5. Distribution-free methods

Our "for every F" results sit in the tradition of Scarf [1958] and Gallego and Moon [1993]. Perakis and Roels [2008] formalizes what is and is not decidable under partial distributional information, which is the right frame for our conditional-mean dependence. We deliberately do *not* claim a distributionally robust optimization result: a referee would reasonably ask why we do not simply solve a DRO problem over F [Delage and Ye, 2010, Wiesemann et al., 2014], and our worst-case guarantee has infimum zero, so the DRO vocabulary would promise more than we deliver. On detail-free design, our contract is simple because the designer will not commit to a distribution, in the spirit of Wilson [1987] and Bergemann and Morris [2005]; Carroll [2015] is the model for how to make that argument, and we note that our setting has *no private information*, so we avoid mechanism-design language entirely.

6. Carbon accounting

Φ is built on conditional means, which is marginal-emissions-factor logic done exactly; Siler-Evans et al. [2012] established that average factors badly misestimate avoided emissions. Two boundaries of our accounting deserve statement. Our counterfactual is *no load*, but a curtailed miner relocates rather than vanishing, so grid-local accounting understates leakage [Fowlie, 2009]; and our unconstrained-rule result is a supply-side backfire of the kind catalogued by Gillingham et al. [2016]. Bushnell and Novan [2021] further suggests our two-conditional-mean summary may be too coarse to capture ramping and start-up effects.

7. Summary of the contribution

Against this literature we claim: (i) a tractable joint capacity characterization with distribution-free cutoffs and a recursive-triangular uniqueness argument; (ii) an exact closed-form carbon ledger on the large-FCL regime; and (iii) the price-versus-quantity comparison of §5.3—allocation equivalence at the greening thresholds, revenue divergence, and a strict long-run ranking—which we have not found anywhere in this literature. We do *not* claim the direction of the naive-rule carbon effect, which is Knittel et al. [2025] and Holland and Mansur [2008].

References

Sam Aflaki and Serguei Netessine. Strategic investment in renewable energy sources: The effect of supply intermittency. *Manufacturing & Service Operations Management*, 19(3):489–507, 2017. doi: 10.1287/msom.2017.0621.

Ross Baldick, Sergey Kolos, and Stathis Tompaidis. Interruptible electricity contracts from an electricity retailer's point of view: Valuation and optimal interruption. *Operations Research*, 54 (4):627–642, 2006.

Dirk Bergemann and Stephen Morris. Robust mechanism design. *Econometrica*, 73(6):1771–1813, 2005. doi: 10.1111/j.1468-0262.2005.00638.x.

James Bushnell and Kevin Novan. Setting with the sun: The impacts of renewable energy on conventional generation. *Journal of the Association of Environmental and Resource Economists*, 8(4):759–796, 2021. doi: 10.1086/713249.

R. Andrew Butters, Jackson Dorsey, and Gautam Gowrisankaran. Soaking up the sun: Battery investment, renewable energy, and market equilibrium. *Econometrica*, 93(3):891–927, 2025. doi: 10.3982/ECTA20411.

Duncan S. Callaway, Meredith Fowlie, and Gavin McCormick. Location, location, location: The variable value of renewable energy and demand-side efficiency resources. *Journal of the Association of Environmental and Resource Economists*, 5(1):39–75, 2018. doi: 10.1086/694179.

Gabriel Carroll. Robustness and linear contracts. *American Economic Review*, 105(2):536–563, 2015. doi: 10.1257/aer.20131159.

Richard T. Carson and Kevin Novan. The private and social economics of bulk electricity storage. *Journal of Environmental Economics and Management*, 66(3):404–423, 2013.

Hung-po Chao and Robert Wilson. Priority service: Pricing, investment, and market organization. *American Economic Review*, 77(5):899–916, 1987.

Hung-po Chao, Shmuel S. Oren, Stephen A. Smith, and Robert B. Wilson. Priority service: Market structure and competition. *The Energy Journal*, 9:77–104, 1988. Special Issue on Electricity Reliability.

Michael A. Crew and Paul R. Kleindorfer. Peak load pricing with a diverse technology. *Bell Journal of Economics*, 7(1):207–231, 1976.

Erick Delage and Yinyu Ye. Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Operations Research*, 58(3):595–612, 2010. doi: 10.1287/opre.1090.0741.

Yongbin Ding, Mingquan Li, Ahmed Abdulla, Rui Shan, and Ziyi Liu. Unintended consequences of curtailment cap policies on power system decarbonization. *iScience*, 26(7):106967, 2023.

David F. Drake, Paul R. Kleindorfer, and Luk N. Van Wassenhove. Technology choice and capacity portfolios under emissions regulation. *Production and Operations Management*, 25(6):1006–1025, 2016. doi: 10.1111/poms.12523.

Ali Fattahi, Sriram Dasu, and Reza Ahmadi. Peak-load energy management by direct load control contracts. *Management Science*, 69(5):2788–2813, 2023. doi: 10.1287/mnsc.2022.4493.

Meredith L. Fowlie. Incomplete environmental regulation, imperfect competition, and emissions leakage. *American Economic Journal: Economic Policy*, 1(2):72–112, 2009. doi: 10.1257/pol.1.2.72.

Guillermo Gallego and Ilkyeong Moon. The distribution free newsboy problem: Review and extensions. *Journal of the Operational Research Society*, 44(8):825–834, 1993. doi: 10.1057/jors.1993.141.

Kenneth Gillingham, David S. Rapson, and Gernot Wagner. The rebound effect and energy efficiency policy. *Review of Environmental Economics and Policy*, 10(1):68–88, 2016. doi: 10.1093/reep/rev017.

Eric S. Hittinger and Inês M. L. Azevedo. Bulk energy storage increases united states electricity system emissions. *Environmental Science & Technology*, 49(5):3203–3210, 2015. doi: 10.1021/es505027p.

Stephen P. Holland and Erin T. Mansur. Is real-time pricing green? the environmental impacts of electricity demand variance. *Review of Economics and Statistics*, 90(3):550–561, 2008.

Shanshan Hu, Gilvan C. Souza, Mark E. Ferguson, and Wenbin Wang. Capacity investment in renewable energy technology with supply intermittency: Data granularity matters! *Manufacturing & Service Operations Management*, 17(4):480–494, 2015. doi: 10.1287/msom.2015.0536.

Rajnish Kamat and Shmuel S. Oren. Exotic options for interruptible electricity supply contracts. *Operations Research*, 50(5):835–850, 2002.

Christopher R. Knittel, Juan Ramon L. Senga, and Shen Wang. Flexible data centers and the grid: Lower costs, higher emissions? NBER Working Paper 34065, National Bureau of Economic Research, 2025. Published as "Flexible Data Centers Reduce Power System Costs but Can Increase Emissions," iScience (2026), art. 116497. Cite the NBER version; the iScience DOI did not resolve cleanly at time of writing.

A. Gürhan Kök, Kevin Shang, and Şafak Yücel. Impact of electricity pricing policies on renewable energy investments and carbon emissions. *Management Science*, 64(1):131–148, 2018. doi: 10.1287/mnsc.2016.2576.

A. Gürhan Kök, Kevin Shang, and Şafak Yücel. Investments in renewable and conventional energy: The role of operational flexibility. *Manufacturing & Service Operations Management*, 22(5):925–941, 2020.

Zhenhua Liu, Minghong Lin, Adam Wierman, Steven H. Low, and Lachlan L. H. Andrew. Greening geographical load balancing. In *Proceedings of ACM SIGMETRICS 2011*, pages 233–244, 2011. doi: 10.1145/1993744.1993767.

Ali Menati, Kiyeob Lee, and Le Xie. Modeling and analysis of utilizing cryptocurrency mining for demand flexibility in electric energy systems: A synthetic texas grid case study. *IEEE Transactions on Energy Markets, Policy and Regulation*, 2023a. IEEE Xplore document 9999040; arXiv:2207.02428. Volume/issue/pages unconfirmed.

Ali Menati, Xiangtian Zheng, Kiyeob Lee, Ranyu Shi, Pengwei Du, Chanan Singh, and Le Xie. High resolution modeling and analysis of cryptocurrency mining's impact on power grids: Carbon footprint, reliability, and electricity price. *Advances in Applied Energy*, 10:100136, 2023b.

Kevin Novan. Valuing the wind: Renewable energy policies and air pollution avoided. *American Economic Journal: Economic Policy*, 7(3):291–326, 2015. doi: 10.1257/pol.20130268.

Shmuel S. Oren and Stephen A. Smith. Design and management of curtailable electricity service to reduce annual peaks. *Operations Research*, 40(2):213–228, 1992. doi: 10.1287/opre.40.2.213.

Xiaoshan Peng, Owen Q. Wu, and Gilvan C. Souza. Renewable, flexible, and storage capacities: Friends or foes? *Manufacturing & Service Operations Management*, 26(5):1730–1749, 2024. doi: 10.1287/msom.2023.0068.

Georgia Perakis and Guillaume Roels. Regret in the newsvendor model with partial information. *Operations Research*, 56(1):188–203, 2008. doi: 10.1287/opre.1070.0486.

Asfandyar Qureshi, Rick Weber, Hari Balakrishnan, John Guttag, and Bruce Maggs. Cutting the electric bill for internet-scale systems. In *Proceedings of ACM SIGCOMM 2009*, 2009. doi: 10.1145/1592568.1592584.

Ana Radovanović, Ross Koningstein, Ian Schneider, Bokan Chen, et al. Carbon-aware computing for datacenters. *IEEE Transactions on Power Systems*, 38(2):1270–1280, 2023.

Iegor Riepin and Tom Brown. On the means, costs, and system-level impacts of 24/7 carbon-free energy procurement. *Energy Strategy Reviews*, 54:101488, 2024.

Herbert Scarf. A min-max solution of an inventory problem. In K. J. Arrow, S. Karlin, and H. Scarf, editors, *Studies in the Mathematical Theory of Inventory and Production*, pages 201–209. Stanford University Press, 1958.

Kyle Siler-Evans, Inês Lima Azevedo, and M. Granger Morgan. Marginal emissions factors for the u.s. electricity system. *Environmental Science & Technology*, 46(9):4742–4748, 2012. doi: 10.1021/es300145v.

Todd P. Strauss and Shmuel S. Oren. Priority pricing of interruptible electric service with an early notification option. *The Energy Journal*, 14(2):175–195, 1993. doi: 10.5547/ISSN0195-6574-EJ-Vol14-No2-9.

Nur Sunar and John R. Birge. Strategic commitment to a production schedule with uncertain supply and demand: Renewable energy in day-ahead electricity markets. *Management Science*, 65(2):714–734, 2019.

Sam van der Jagt, Neha Patankar, and Jesse D. Jenkins. Understanding the role and design space of demand sinks in low-carbon power systems. *Energy and Climate Change*, 5:100132, 2024. doi: 10.1016/j.egycc.2024.100132.

Jan A. Van Mieghem. Capacity management, investment, and hedging: Review and recent developments. *Manufacturing & Service Operations Management*, 5(4):269–302, 2003. doi: 10.1287/msom.5.4.269.24882.

Adam Wierman, Zhenhua Liu, Iris Liu, and Hamed Mohsenian-Rad. Opportunities and challenges for data center demand response. In *International Green Computing Conference (IGCC)*, pages 1–10, 2014.

Wolfram Wiesemann, Daniel Kuhn, and Melvyn Sim. Distributionally robust convex optimization. *Operations Research*, 62(6):1358–1376, 2014. doi: 10.1287/opre.2014.1314.

Robert Wilson. Game-theoretic analyses of trading processes. In Truman Bewley, editor, *Advances in Economic Theory: Fifth World Congress*, chapter 2, pages 33–70. Cambridge University Press, 1987.